

MetaSTC: Context-Adaptive Meta-Learning for Traffic Forecasting on Large-Scale Road Networks

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Abstract—Large-scale traffic forecasting must handle both persistent spatial heterogeneity and time-varying local regimes. The conference version of MetaSTC addresses the former by clustering roads according to spatio-temporal representations and learning task-specific predictors through meta-learning. However, a fixed task assignment cannot explicitly respond when the recent traffic state deviates from its usual regime. In this journal extension, we preserve the trained MetaSTC experts as a reliable static anchor and introduce a context-conditioned gated residual meta-adapter. The adapter encodes the observed 12-step history together with the static 6-step forecast into a continuous context state, combines it with the static task prior, and predicts a bounded gated correction. Its correction head is zero-initialized so that the extended model exactly recovers the static anchor at initialization, while a safety-aware objective discourages harmful adaptation. Under the official LSTM $L = 12$ protocol, five-seed averages improve the published MetaSTC MAE/MSE on Beijing, Shanghai, and LargeST by 0.65/0.79%, 5.72/7.25%, and 4.00/1.96%, respectively; all 15 dataset-seed MAE results beat the published conference targets. The adapter adds only about 10.9K trainable parameters in absolute terms and introduces no measurable material inference-latency penalty in our batched GPU benchmark. Matched-control and segment-level analyses further show that richer context conditioning is setting-dependent rather than universally superior to simpler residual calibration, with Shanghai providing a clear case where adaptation becomes more beneficial on difficult windows.

Index Terms—Traffic Flow Prediction, Meta-Learning, Spatio-Temporal Heterogeneity, Context-Conditioned Adaptation, Residual Learning, Large-Scale Road Networks.

I. INTRODUCTION

THE traffic flow prediction problem is a crucial endeavor within the field of transportation engineering, focusing on the prediction of temporal sequences that represent future traffic conditions. As illustrated in Figure 1, traffic flow prediction plays a pivotal role in optimizing the functioning of transportation systems such as traffic management strategies, congestion mitigation efforts, and efficient scheduling practices [16].

The field of traffic flow prediction dates back to 1959 by H Greenberg [8]. Over the years, there has been continuous

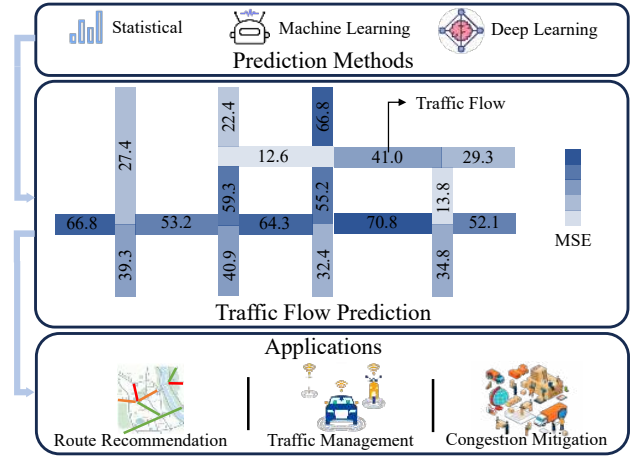


Fig. 1: The Process of Traffic Flow Prediction.

research in the field of traffic prediction, covering a wide range of methods including statistical approaches, traditional machine learning techniques, and more recently, deep learning models. However, existing works face challenges in terms of precision and applicability to **real-world datasets**. We outline these challenges as follows:

Challenge 1: Complex Traffic Mode Traffic flow exhibits inherent time dependence with cyclic fluctuations occurring on daily, weekly, and yearly scales. Spatial disparities can be ascribed to various factors such as differences in economic development levels, area topography, and planning strategies. Figure 2a shows the differences in traffic patterns for three roads from different locations. The flow sequences of the three roads can be seen to show a daily regularity but with different patterns of increase and decrease. Road 2 shows a peak during the day and decreases in flow as the night goes on, while the remaining two do the opposite. The mean values of flows for the first road and the second road are close to and higher than the rest. It is difficult to fit all the roads in the network with the same set of parameters.

Challenge 2: Large Scale Dataset Due to the inextricable relationship between the traffic prediction problem and real-world scenarios, We should consider the feasibility and effectiveness of model design applied to large-scale datasets of real scenarios. In reality, a mid-sized city requires traffic forecasting including approximately hundreds of service areas and thousands of roadways for each area [14]. Figure 2b demonstrates the relationship between the monitoring data

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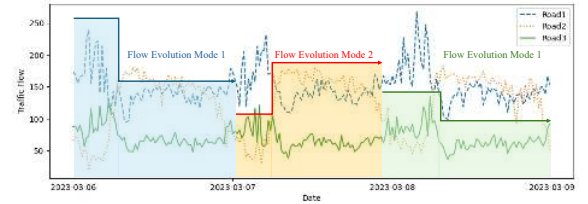
used for actual downstream forecasting and the public datasets used for traffic forecasting research. The current test results on previous small public datasets are not necessarily applicable to larger datasets. Test results on large-scale public data [14] have also shown that previous work has degraded performance on large-scale datasets. The performance of the model in the face of large-scale datasets is yet to be explored.

Challenge 3: Diverse Spatio-Temporal Scenarios For different road scenarios, the traffic flow pattern laws show different data distributions. For example, in metropolitan areas, traffic near business districts will have significant peaks in commuting time intervals while these traffic modes do not apply in the countryside. Figure 2c shows different scenarios such as cities, villages, campuses, and resorts. Under different data scenarios, the performance of one model varies greatly, and it is difficult to have a generalized model to fit all the datasets with different data distributions [18].

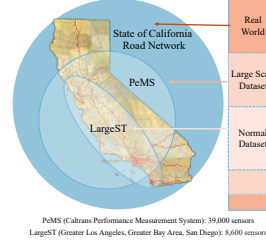
The conference MetaSTC formulation addresses persistent heterogeneity by assigning roads to spatio-temporal tasks and learning task-specific predictors. A remaining limitation is that traffic regimes evolve over time: even a road with a stable task identity may encounter a recent context that differs from the conditions represented by its static expert. Re-clustering or retraining the entire predictor whenever the regime changes is undesirable for a large road network. The journal extension therefore separates a reliable *static anchor* from a lightweight *dynamic correction*. The static MetaSTC expert retains the conference model’s spatial/task prior, while a continuous context state extracted from recent observations decides whether and how much residual correction should be applied. This design preserves the established conference model and makes adaptation conservative rather than replacing the predictor with a new routing mechanism.

To address the aforementioned challenges, we develop a **Meta Spatio-Temporal Clustering Learning Paradigm** named **MetaSTC**. The conference backbone contains a spatio-temporal representation clustering module and a spatio-temporal meta-learning module; the journal extension further equips this backbone with a context-conditioned gated residual meta-adaptor for time-varying local regimes.

Gabriel Cirstea et al. [4] mentioned that a shared network for all the road networks forces the learned parameters to represent an “average” of the traffic patterns among all traffic flow series. Considering the complex traffic mode in road network (mentioned in **Challenge 1**), they used spatial-aware modeling where different model parameters can be used for modeling time series from different locations. But fitting every feature pattern on the road network with a separate parameter again brings excessive training time as well as the problem of overfitting. In order to reduce the variability of feature distribution in the dataset, we include K-Means method to group together roads into clusters. Previous clustering based methods for traffic prediction (e.g. Hwang Kuo et al. [11]) lack the discussion of spatio-temporal features. We design a spatio-temporal gating unit to capture spatio-temporal contextual information and apply the clustering based on obtained spatio-temporal representation. For the road nodes within the same cluster, the differences between them are relatively small



(a) Complex ST Correlations



(b) Large Scale Dataset



(c) Diverse Scenarios

Fig. 2: Challenges of Traffic Flow Prediction.

because they share similar spatio-temporal patterns.

As mentioned above, state-of-the-art models have demonstrated exceptional performance on small-scale datasets (e.g. Pemsd [3]) by considering the dynamic characteristics of spatial topology, but their complex designs prevent them from scaling to larger datasets [14]. More complex network structures require higher computational resources (e.g. running memory, training time) and limit the size of the batch size with the same computational resources, which brings about instability in training. To enhance the applicability of the model to real-world datasets (**challenge 2**), we design a meta-learner that refines global models with specific subtasks rather than training a mega-scale model. The meta-learner makes it possible to represent task-specific details with a simpler model, and reduces the likelihood of its overfitting. Meta-learning can be seen as an advanced form of transfer learning that transfers not only knowledge but also learning strategies to improve overall stability and performance through information sharing between tasks [7].

Previous work applying meta-learning to solve traffic flow prediction problems [17] [26] [21] lacked experimental validation on large-scale datasets and didn’t put any restrictions on choosing subtasks. We improved the meta-learner by developing a contextual memory unit. Building upon the earlier established categories, our contextual memory unit extracts sub-task-related information for model prediction. The meta-learner learns information from the single road scope, from the subtask scope, and from the entire dataset scope which improves the prediction performance on large-scale datasets.

In order to cope with the great variability of traffic flow pattern laws in different road scenarios (**challenge 3**), our framework adopts backbone agnostic modularity design, which allows key components such as the prediction backbone network, feature fusion module, etc. to be independently upgraded or replaced, and facilitates the rapid iteration of the model structure according to the scenarios.

In all, our contributions can be summarized as follows.

TABLE I: Advantages of MetaSTC compared to other traffic flow prediction methods.

Advantages	Spatio-Temporal Correlation	Divide-and-Rule	Backbone Agnostic
TimesNet [22]	×	×	×
STGCN [24]	✓	×	×
ASTGCN [30]	✓	×	×
AGCRN [1]	✓	×	×
PDFormer [10]	✓	×	×
DCRNN [13]	✓	×	×
Graph Wavenet [23]	✓	×	×
ST-WA [4]	✓	×	✓
MetaSTC(ours)	✓	✓	✓

- 1) We retain MetaSTC as a scalable, backbone-agnostic meta-learning framework for large road networks. Spatio-temporal representation clustering captures persistent heterogeneity and provides a strong static task prior instead of fitting one global parameter set to all roads.
- 2) As the main journal extension, we introduce a *context-conditioned gated residual meta-adapter*. It combines a continuous state representation of the recent traffic history and static forecast with the MetaSTC task prior, and learns only a bounded residual correction. Zero initialization makes the extended model exactly recover the static MetaSTC anchor before adaptation, providing a conservative inheritance path from the conference model.
- 3) We evaluate the mechanism with matched residual controls and segment-level diagnostics rather than assuming that a more complex adapter is always necessary. The analyses identify when richer context conditioning is useful and expose settings in which a simple residual calibrator is competitive or stronger, leading to calibrated rather than universal mechanism claims.
- 4) We provide a five-seed journal evaluation on Beijing, Shanghai, and LargeST under the official LSTM $L = 12$ setting. V14 improves the published MetaSTC MAE on every dataset and every tested seed, improves the five-seed MSE mean on all three datasets, adds only about 10.9K trainable parameters, and shows no measurable material inference-latency penalty under batched GPU evaluation.

II. PRELIMINARIES AND PROBLEM FORMULATION

In this section, we first define the key concepts for modeling the flow of traffic on the road and then formulate our problem.

Traffic Flow Sequence We define the traffic network as a directed graph $G = (V, E)$, where V is the set of nodes that represents the intersections of the roads, and E is the set of edges that represents the roads. At a specific time stamp t , a traffic flow sequence X_t represents a L long historical traffic flow data on the entire road network. Specifically, $X_t = \{x_{t-L}, \dots, x_{t-1}\}$, where x_i represents the traffic flow at the historical time stamp i for all roads. In our discussion, we use the speed of the traffic to represent the traffic flow.

Traffic Flow Prediction Problem Given time stamp t , the traffic flow prediction problem aims to predict the future traffic flow for all roads. More specifically, provided

the traffic flow sequence X_t , a prediction model p predicts the traffic flow for the next few time stamps $p(X_t) = \{\hat{x}_{t+1}, \dots, \hat{x}_{t+P}\} = \{\hat{x}\}_{t+1}^{t+P}$, where P denotes the prediction horizon. The goal is to minimize the cumulative loss $L = \sum_{t=L+1}^T \mathcal{L}(\{\hat{x}\}_{t+1}^{t+P}, \{x\}_{t+1}^{t+P})$, where T is the total number of predicted steps, and $\mathcal{L}(\cdot, \cdot)$ can be any loss function such as MAE and RMSE.

III. METHODOLOGY

Our MetaSTC paradigm is composed of two major modules: the road clustering module and the meta traffic forecasting module. Our overall framework is outlined in Figure 3. The clustering module reduces the variability of spatio-temporal patterns in the same subtask and meta-learning can adaptively adjust training for different subtasks. The proposed flow forecasting model is customized for each cluster, i.e., the parameters in the neural network are the same for the roads within the same cluster and vice versa.

A. Spatio-Temporal Representation Clustering

As **challenge 1** describes, traffic flow inherently possesses temporal and spatial dependencies. In order to better capture the cross-region traffic flow transfer patterns, we propose a spatio-temporal representation learning method for roads. In addition to the dynamic traffic flow patterns, we consider as well as static attributes of road segments that influence the actual travel flow such as the road index (roadID), road level (roadLevel), and the number of lanes (laneNum). Table II depicts all the spatio-temporal features in MetaSTC. We train a gating mechanism to embed the dynamics and interactions amongst roads. Finally, we cluster roads with similar patterns.

Temporal Features To capture the various changing temporal patterns mentioned above, we start with a standardized approach to reprocess traffic flow data. Standardization helps to remove the scale and shift differences that might exist in the data, making it more suitable for further analysis.

As mentioned in the introduction, traffic flow data exhibit strong periodicity on a day and week scale. Recognizing these clear cyclical trends is critical for accurate traffic prediction and analysis. Therefore, we encode a time point t with a combination of linear and periodic functions that encapsulate these two essential temporal dimensions. The encoding is formulated as

$$\text{Traffic2Vec}(X_t) = [\text{Traffic2Vec}(t) : \text{Traffic2Vec}(t + T)] \quad (1)$$

$$\text{Traffic2Vec}(t) = [w_0 t + b_0, \sin(w_d t + \phi_1), \sin(w_w t + \phi_2)]. \quad (2)$$

Here, $w_0 t + b_0$ represents the linear component and $\sin$ terms encode the daily and weekly cycles respectively, where w_d denotes for the daily frequency ($\frac{2\pi}{24}$) and w_w denotes for the weekly frequency ($\frac{2\pi}{7 \times 24}$). The ϕ terms are learnable parameters that denote phase shifts, allowing our model to adjust to the specific temporal dynamics of the traffic flow. After computing the temporal encoding, we apply a dense layer, to further process the sequence

$$h_t = \text{Dense}(X_t, \text{Traffic2Vec}(X_t)) \quad (3)$$

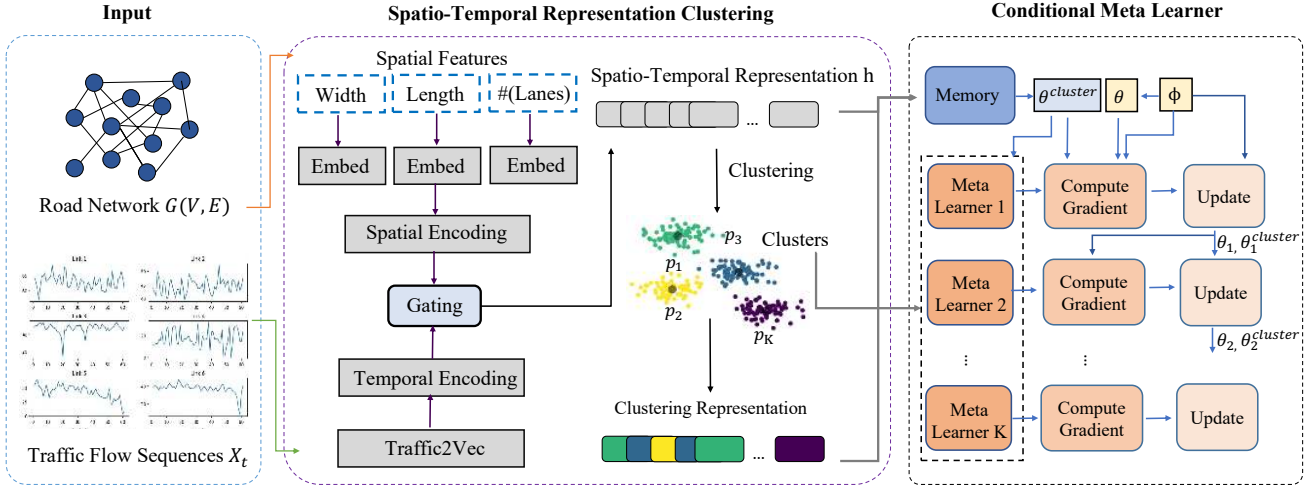


Fig. 3: The Framework of Metastsc. Our model contains two components: a spatio-temporal representation clustering module and a meta-learning module.

TABLE II: Spatial and Temporal Features.

	Feature	Description	Possible Range of Values
Temporal	MonthID	An identifier for each month in a year	Integer from 1 to 12
	WeekID	An identifier for each day of the week	Integer from 1 to 7
Spatial	RoadID	The identifier of the sensor	A number with 6 to 9 digits
	RoadLevel	The hierarchical level of road	Primary, Secondary, Tertiary
	Lat	The latitude of road	Real number between -90 and 90
	Lng	The longitude of road	Real number between -180 and 180
	LaneNum	The number of lanes	Integer from 1 to 8
	Width	The width of road	Real number (unit: meters)
	Length	The length of road	Real number (unit: meters)
	Direction	The direction of the highway	North (N), South (S), East (E), West (W)

where Dense represents the dense layer with its own set of learnable weights. By constructing the temporal hidden state in this way, the model can capture both the explicit temporal encoded patterns and the implicit patterns which increases the interpretability of models.

Spatial Features Spatial attributes of road segments also play an important role in traffic flow prediction. Part of the attributes, such as the road index, road level, and the number of lanes, are categorically valued and thus cannot be fed into the model directly. It is generally computationally expensive to tackle these features by the one-hot encoding. To solve this problem, we embed the discrete features into a low-dimensional space to seek compact representations as well as to share similar patterns across different road segments. The embeddings are concatenated to form the spatial hidden state

$$h_s = M_{index} \oplus M_{level} \cdots \oplus M_{lane} \quad (4)$$

where $M_{index}, M_{level}, M_{lane}$ are trainable embedding matrices and $\oplus$ represents the concatenate operation.

Spatio-Temporal Gating Unit The integration of spatial and temporal features in traffic flow prediction requires a mechanism capable of modeling their complex, nonlinear interactions. Simply concatenating or adding these features, as

done in traditional approaches, risks losing critical contextual information due to the heterogeneous nature of traffic data across time and space. In our context, the spatio-temporal gating unit is designed to dynamically weigh the contributions of spatial and temporal hidden states based on their relevance to the prediction task. This approach aligns with the principle of attention in sequence modeling, where the model learns to focus on salient features. Mathematically, the gating vectors G_s and G_t (defined below) act as adaptive filters, modulating the influence of h_s and h_t through a sigmoid activation. This ensures that the final representation h preserves both spatial topology and temporal periodicity while mitigating noise from irrelevant features, a critical requirement given the cyclic yet variable nature of traffic patterns (Challenge 1).

We want to find a way to transform the spatial and temporal features into a format that captures the essential spatio-temporal context information. Instead of simply adding or concatenating, here we design a gating unit linking spatial and temporal representations to obtain the hidden states used for clustering. This allows the model to control the flow of information more flexibly by learning weights to understand complex nonlinear relationships to better capture the interactions between different features. As a result, the model

can automatically determine the level of contribution of each hidden state based on a specific task, which is more flexible and adaptable. The formulas are as follows

$$G_s = \text{sigmoid}(W_{ss}h_s + W_{ts}h_t + b_s) \quad (5)$$

$$G_t = \text{sigmoid}(W_{st}h_s + W_{tt}h_t + b_t) \quad (6)$$

$$h = [G_s \odot h_s || G_t \odot h_t]. \quad (7)$$

Here, G_s and G_t are gating vectors for spatial and temporal states, respectively. W_{ss} and W_{tt} are the self-gating weight matrices for the spatial and temporal states. In contrast, W_{ts} and W_{st} are cross-gating weight matrices that allow each state to influence the gating of the other. The biases b_s and b_t adjust the activation of the gates.

The final hidden state h combines the gated spatial and temporal states, preparing the features for clustering and further analysis. This unit enables the model to dynamically prioritize salient spatio-temporal features for improved predictive performance. We train all the parameters during the global updating process of the flow forecasting task.

From a signal processing perspective, the gating unit can be viewed as a dynamic filter that disentangles the periodic components of traffic flow (e.g., daily and weekly cycles) from spatially-driven variations (e.g., road hierarchy and topology). This is particularly advantageous in traffic data, where temporal periodicity is often corrupted by spatial heterogeneity, such as differing peak hours across urban and rural roads (Challenge 1). By assigning learnable cross-gating weights W_{ts} and W_{st} , the model effectively performs a form of feature selection, amplifying relevant spatio-temporal interactions while suppressing noise, akin to adaptive filtering in time-series analysis. This theoretical underpinning ensures that the resulting representation h is not only expressive but also robust to the multi-scale dependencies inherent in large-scale traffic networks.

Spatio-Temporal Road Network Clustering As is mentioned in **challenge 1**, traffic flow patterns vary greatly from time to time and from location to location, while a unified model uses a single set of parameters over all the roads, and thus has limits in capturing data variability over large regions. Therefore, we thought of training multiple sub-regions separately. We conduct K-Means++ over the spatio-temporal embeddings by the representation learning model to obtain the clustering over the road network.

The process of the representation clustering module is described in Alg. 1. When encountering a new road, we can rapidly obtain representation and assigned tasks.

B. Spatio-Temporal Meta Learner

Traffic flow prediction on large-scale datasets demands a model that can generalize across diverse road segments while adapting to their unique spatio-temporal characteristics. Traditional deep learning models, trained on a single global parameter set, often struggle to capture this heterogeneity, resulting in suboptimal performance on unseen scenarios (Challenge 2). To overcome this, we adopt a meta-learning paradigm, rooted in the concept of “learning to learn”, which enables rapid adaptation to new tasks by leveraging prior knowledge. Unlike

Algorithm 1: Spatio-Temporal Representation Clustering

Input: Traffic flow sequences X_t , Road features f_s , Number of clusters K

Output: Spatio-Temporal representation h , Cluster centroids $p_1, p_2, \dots, p_K$

```

1 foreach  $x_{i,t} \in X_t$  do
2    $h_{i,s} \leftarrow \text{Spatio Encoder}(f_{i,s});$ 
3    $h_{i,t} \leftarrow \text{Temporal Encoder}(x_{i,t});$ 
4    $h_i \leftarrow \text{Gating}(h_{i,s}, h_{i,t});$ 
5 end
6 Initialize  $p_1, p_2, \dots, p_K$  at random;
7 while not converged do
8   foreach  $x_{i,t} \in X_t$  do
9     for  $j = 1$  to  $K$  do
10       $\mathbb{P}_j(i) \leftarrow \frac{\text{Distance}(h_i - p_j)^2}{\sum \text{Distance}(h_i - p_j)^2};$ 
11    end
12  end
13  Sample  $p_1, p_2, \dots, p_K$  from  $X \sim \mathbb{P}(X);$ 
14 end

```

standard transfer learning, meta-learning optimizes an initial parameter set that can be fine-tuned efficiently for specific subtasks—in our case, clusters of roads with similar patterns. This approach is theoretically grounded in Model-Agnostic Meta-Learning (MAML) [7], where a meta-learner minimizes the expected loss across tasks via second-order optimization. By integrating clustering with meta-learning, we ensure that the model not only captures global traffic dynamics but also tailors predictions to local contexts, enhancing scalability and robustness on real-world datasets. The subsequent design of our cluster-aware memory unit builds upon this foundation, embedding task-specific priors into the learning process.

As **challenge 2** mentioned, due to the inextricable relationship between the traffic prediction problem and real-world scenarios, models that can be applied to large-scale datasets of real-world scenarios are more meaningful. To address this, we proposed a spatio-temporal meta-learning module that aims to better adapt the prediction model to large-scale datasets. In urban areas, the temporal patterns, characteristics of locations, and the mutual relationship of roads are diverse. So far, we have divided the entire road network into multiple categories of smaller sizes based on the spatio-temporal features of each road. In this spatio-temporal meta-learning module, we use the meta-learner to train unique model parameters for each category. The separation of learning of category priors makes it possible to represent road-specific details with a simpler model within the smaller dataset. Each time we initial from the global parameters based on the previous pre-trained tasks.

The synergy of clustering and meta-learning in our framework further optimizes the bias-variance trade-off, a key concern in statistical learning for heterogeneous datasets. By partitioning the road network into clusters with similar spatio-temporal patterns, we reduce the variance within each subtask, allowing the meta-learner to focus on task-specific adaptations rather than fitting a single, overgeneralized model.

This aligns with the principle of task distribution modeling in meta-learning, where the meta-learner implicitly learns a prior over a family of related tasks (i.e., clusters) rather than individual data points. Consequently, the model achieves improved generalization on unseen roads by leveraging shared knowledge across clusters, while the localized fine-tuning mitigates overfitting—a critical advantage when scaling to thousands of roads with diverse distributions (Challenge 2).

Cluster-aware Memory Unit To accommodate the unique characteristics of roads influenced by various contextual factors, it is impractical to use a one-fits-all approach with global parameters for traffic flow prediction. Recognizing the impact of road diversity, we have formulated a cluster-aware contextual memory unit M_c that preserves distinct parameters for the predictive layer tailored to different road clusters. This memory unit is responsible for dispensing personalized initial parameters for each road, drawing on a similarity score computed by a prescribed algorithm.

Upon completion of a clustering operation, we ascertain the centroid of each cluster p_j , with j indexing the clusters. We then define a similarity score, represented as a to quantify the closeness of a given road's features to the cluster centroid. The similarity score a_j^k between road j and cluster k can then be recalculated using a softmax function as

$$a_j^k = \frac{\exp(p_k \cdot h_j)}{\sum_{i=1}^K \exp(p_i \cdot h_j)} \quad (8)$$

where K represents the total number of cluster and h_j reflects the hidden state of j -th road calculated by equation 7. Specifically, we retrieve the parameter matrix $M_{j,c}$ for the j -th road $M_{j,c} = a_j^k \cdot M_c$.

In the spirit of LSTM [9], our memory unit integrates a reset gate and an update gate to manage its parameters effectively. As the training process unfolds, the memory unit retrieves and updates the initial parameters M_c as

$$r_t = \sigma(W_r \cdot (a_j^k \otimes M_{j,c}) + b_r) \quad (9)$$

$$u_t = \sigma(W_u \cdot (a_j^k \otimes M_{j,c}) + b_u) \quad (10)$$

$$\tilde{M}_c = \tanh(W_c \cdot (r_t \odot (a_j^k \otimes M_{j,c})) + b_c) \quad (11)$$

$$M_c = (1 - u_t) \otimes M_c + u_t \otimes \tilde{M}_c \quad (12)$$

In these equations, r_t is the reset gate at time t , u_t the update gate, h_{t-1} the parameter state at the previous time step, and σ denotes the sigmoid activation function. The weight matrices W_r , W_u , and W_c , along with their corresponding bias terms b_r , b_u , and b_c , adjust the gates and parameter updates accordingly. M_c is then utilized by the meta learner to initialize the local parameters at the current timestep. This results in a contextually informed and adaptive prediction model capable of yielding more accurate and reliable traffic flow estimations.

Local Update In traditional meta model such as MAML [7], we split the traffic flow sequences into a training set $\mathcal{D}_{train}$ and a testing set $\mathcal{D}_{test}$ and partition the training set as the support set and query set. The support set is to adapt the model's parameters, and the query set is to evaluate how well the adapted model can generalize to new data. Our target is to find optimal parameters such that for each specific task, it produces task-specific parameters on all the tasks and helps

the meta-model achieve the lowest error on their validation data. This can be expressed as

$$\Theta^* = \underset{\Theta}{\operatorname{argmin}} \frac{1}{N} \sum_i \mathcal{L}(\text{meta-model}(\mathcal{D}_{i,test}; \Theta_i), \mathcal{D}_{i,test}). \quad (13)$$

where N is the number of tasks, and Θ denotes all the parameters in our meta-learning model, i denotes the i -th task. We can bring in any model for sequence prediction to combine with our meta-model. We solved **challenge 3** with this backbone agnostic design. During the experiments, we tested our paradigm by combining LSTM and FiLM models.

On this basis, we design our spatio-temporal meta-learner. Our target is to ensure that initial parameters for two roads are similar if they share similar spatio-temporal embeddings. We initialize clustering parameters $\theta^{cluster}$ from the contextual memory unit M_c that are specific to the clustering component of the meta-model. We also have shared parameters θ that are shared across all roads. We initialize the local parameters θ from the global parameters Θ . They capture the common features and patterns that are relevant for all roads. They help improve the prediction accuracy of unique tasks which equals categories in our paradigm. The local learning rate is denoted as β . We rewrite the local updating process of our meta-learning process as

$$\theta_i^{cluster} \leftarrow \theta_i^{cluster} - \beta \nabla_{\theta_i^{cluster}} \mathcal{L}_{D_s}, \quad (14)$$

$$\theta_i \leftarrow \theta_i - \beta \nabla_{\theta_i} \mathcal{L}_{D_s} \quad (15)$$

where $\mathcal{L}_{D_s}$ represents the loss on the support set.

Global Update

$$\Phi \leftarrow \Phi - \gamma \sum \nabla_{\Phi} \mathcal{L}_{D_q} \quad (16)$$

where $\mathcal{L}_{D_q}$ represents the loss function on the query set and the global learning rate is denoted as γ . Both local update and global update are computed via back-propagation. This workflow exhibits high scalability, enabling us to efficiently incorporate new nodes by seamlessly integrating them into meta-tasks and obtaining personalized prediction results.

C. Context-Conditioned Gated Residual Meta-Adaptation

The original MetaSTC experts model persistent heterogeneity among road groups. We use these trained experts as a frozen *static anchor* and adapt only the residual error induced by the current traffic context. Thus, the dynamic component is not a replacement for the static task structure, but a conservative correction on top of an already competitive predictor.

Static anchor and continuous context. For road i belonging to static MetaSTC task c_i , the frozen expert gives a P -step forecast

$$\hat{\mathbf{y}}_{i,t}^{\text{sta}} = f_{c_i}(\mathbf{X}_{i,t}), \quad (17)$$

where $\mathbf{X}_{i,t}$ is the observed L -step history. In the journal experiments $L = 12$ and $P = 6$. We construct a continuous state vector from the normalized history, the normalized static forecast, and local statistics describing recent level, variation, and trend. This gives a 26-dimensional context $\mathbf{s}_{i,t}$ using inference-time information only. A state encoder maps it to

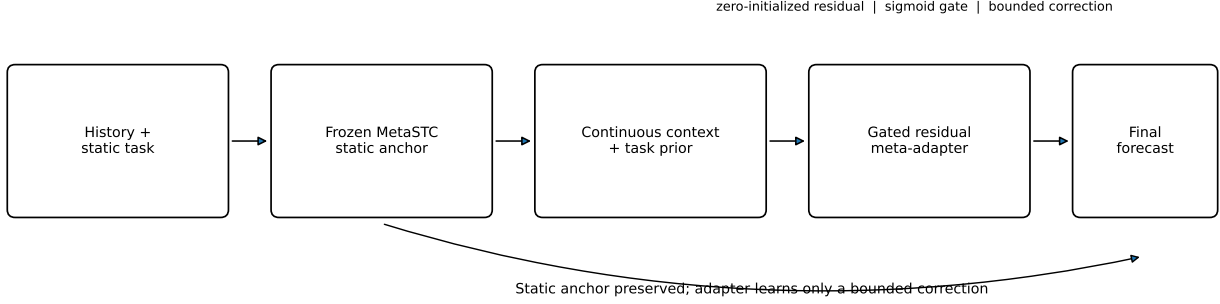


Fig. 4: Journal extension of MetaSTC. A frozen task-specific expert produces the static anchor forecast. Recent history and the anchor are summarized into a continuous context state and combined with the static task prior. A gated residual meta-adapter then applies a bounded correction. The residual branch is zero-initialized, so the extended model exactly recovers the static anchor before adaptation.

$\mathbf{z}_{i,t} = E(\mathbf{s}_{i,t})$. Static task identity is retained through an embedding $\mathbf{e}_{c_i}$,

$$\mathbf{q}_{i,t} = [\mathbf{z}_{i,t} \parallel \mathbf{e}_{c_i}]. \quad (18)$$

The conference task prior therefore remains stable while recent observations provide a continuously varying state representation.

Gated residual modulation. A shared residual branch embeds normalized history and static forecast into $\mathbf{h}_{i,t}$. A lightweight modulator maps $\mathbf{q}_{i,t}$ to feature-wise scale and shift vectors $\gamma_{i,t}$ and $\beta_{i,t}$,

$$\tilde{\mathbf{h}}_{i,t} = \text{GELU}(\mathbf{h}_{i,t} \odot [1 + 0.2 \tanh(\gamma_{i,t})] + 0.2\beta_{i,t}). \quad (19)$$

The adapter predicts a bounded normalized residual and a scalar gate,

$$\delta_{i,t} = \rho \tanh(W_\delta \tilde{\mathbf{h}}_{i,t} + b_\delta), \quad (20)$$

$$g_{i,t} = \sigma(W_g \mathbf{q}_{i,t} + b_g), \quad (21)$$

where ρ bounds the normalized correction. Let $a_{i,t}$ be a local scale estimated from recent history. The final prediction is

$$\hat{\mathbf{y}}_{i,t} = \hat{\mathbf{y}}_{i,t}^{\text{sta}} + a_{i,t} g_{i,t} \delta_{i,t}. \quad (22)$$

The correction head and modulation layer are zero-initialized, making $\delta_{i,t} = \mathbf{0}$ initially. Hence Eq. (22) initially equals the frozen MetaSTC forecast.

Safety-aware optimization. Static experts remain frozen and only the adapter is optimized:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_1 \mathcal{L}_{\text{MAE}} + \lambda_2 [\mathcal{L}_{\text{dyn}} - \mathcal{L}_{\text{sta}}]_+ + \lambda_3 \|\mathbf{r}\|_2^2, \quad (23)$$

where $\mathbf{r} = a_{i,t} g_{i,t} \delta_{i,t}$. The hinge term penalizes samples for which adaptation is worse than the anchor; the correction regularizer discourages unnecessarily large deviations.

IV. EXPERIMENT

In this section, we first introduce the experimental setup including the dataset, running environment, evaluation metrics, baseline methods, and model parameters. Following that, we

will proceed to conduct a series of experiments aimed at addressing the following questions:

- **RQ1:** How does MetaSTC compare with representative traffic forecasting baselines on large-scale real-world datasets?
- **RQ2:** Are the original clustering and meta-learning components necessary, and what structure is captured by learned static tasks?
- **RQ3:** How does the backbone-agnostic MetaSTC framework trade prediction accuracy for computational cost?
- **RQ4:** Does context-conditioned residual adaptation provide stable journal-level improvements over the published MetaSTC results, and how does it compare with a stronger static reproduction and matched residual controls?
- **RQ5:** What is the parameter/inference overhead of the journal adapter, and in which traffic regimes does richer context conditioning provide additional value?

To reproduce our results, we upload our code in an anonymous GitHub repository ¹.

A. Experiment Setting

Datasets We use real traffic datasets of two different cities from an anonymous company ² and two public datasets [3] [14] for experiments. For traffic data, we take every 5 minutes as a time interval and use the average of the traffic speed data recorded during the 5 minutes as the traffic speed value for the period according to actual industrial customs and requirements [18]. Figure 5 presents road networks of various cities. Detailed statistics are presented in Table III.

Environment and Device Setting Our experimental environment is configured with the following specifications: Pytorch 2.10, Python 3.8 (Ubuntu 20.04), and CUDA 11.2. The GPU used for the experiment is the A40 with 48GB

¹<https://github.com/GaoYucen/MetaSTC>

²Because of the double-blind requirement, we hide the company name. This will be made public later.

TABLE III: Statistics of the Datasets.

Dataset	#Roads	Time Range
Beijing	7949	2023/01/01-2023/03/31
Shanghai	5144	2023/06/01-2023/06/30
PeMS04	307	2018/01/01 - 2018/02/28
LargeST	8600	2017/01/01–2021/12/31



Fig. 5: Road Networks of Different Cities. We used company-provided data for the first two cities, as well as two publicly available datasets from California traffic data.

memory and a four-way SLI configuration, while the CPU is an AMD EPYC 7543 32-core Processor with 60 vCPUs. The system has a total of 320GB of memory, with a 25GB system disk and a data disk with 50GB of SSD storage, which can be expanded or contracted as required.

Evaluation Metrics: We use the Mean Absolute Error (MAE), Root mean squared error (RMSE), and the Mean Squared Error (MSE) for the evaluation of accuracy. Our training loss is according to MSE.

Baselines: Considering the limitation of computability as well as the greatness of dataset size, we choose the following models as baselines:

- Autoregressive Integrated Moving Average model (ARIMA): ARIMA is a differential integrated moving average autoregressive model [2].
- LSTM (Long Short-Term Memory): LSTM is a temporal recurrent neural network (RNN). Due to its unique design structure, LSTM is suitable for processing and predicting important events in time series with very long intervals and delays [9].
- Temporal Graph Convolutional Network (T-GCN): T-GCN is a traffic flow prediction model based on graph convolutional neural network. The time-varying graph-structured traffic flow data is well handled by splitting the time nodes and capturing the relationship of roads with a graph network at each time step. [28].
- TimesNet: TimesNet divides complex time changes into multiple intra-periodic and inter-periodic changes, which are embedded into a two-dimensional tensor to realize

adaptive discovery of multi-periodicity [22].

- DLinear: DLinear first decomposes the input into a trend component by a moving average kernel and a remainder component. Then apply linear layers to each component and at last sum them up to get the final prediction [25].
- Frequency improved Legendre Memory model (FiLM): FiLM applies Legendre Polynomials projections to approximate information, uses Fourier projection to remove noise, and adds a low-rank approximation to speed up the calculation process [29].

Parameters: Referring to public traffic prediction datasets [3], we take 5 minutes as a time step and predict the next 6 steps from the previous L steps. L is set to 12 and 24. We divide the experimental training set and test set according to 8 : 2. Based on the trade-offs between convergence speed and accuracy, we use the Adam optimizer with an early stop when converging. The hyperparameter settings are as follows: The learning rate is set to 0.001, and the number of epochs is set to 1000. The batch size is set to 64. The dimension of representation is set to 12. The Euclidean function is chosen as the distance function. For the original Beijing experiments the clustering number is set to 5. In the journal V14 evaluation we preserve the dataset-native frozen task counts used by the trained experts: Beijing uses 5 tasks, while Shanghai and LargeST use 3 tasks. The number of layers is set to 3. The journal extension uses the official LSTM $L = 12$ setting with seeds 42–46; static experts, temporal split, and model-selection protocol are kept fixed across matched residual variants.

B. Experiment Results: RQ1

The MetaSTC model supports combining different traffic flow prediction models. In this section, to balance efficiency and performance, we choose to combine the meta-learning module of our model with the LSTM and FiLM models, respectively. Table IV shows the performance and training time of baseline methods and our proposed MetaSTC model.

In terms of prediction accuracy, our MetaSTC model achieves the best performance when combined with the advanced deep learning model FiLM. CNN-based deep learning models like TimesNet have generally achieved relatively good performance, which may be attributed to the strong learning ability of deep learning models.

Spatial-temporal graph neural network-based models combine graph neural networks with multiple temporal learning networks to capture dynamic properties in spatial and temporal dimensions. As the road network data increases, the model complexity as well as the number of parameters increases. With the same memory resources, we have to reduce the batch size. This makes the training of the model less stable and difficult to converge, resulting in poor performance. Besides, the distribution of roads in our dataset is relatively sparse. This indicates that a large number of roads and the sparse distribution of roads lead to the difficulty of the GNN-based models in learning the relationship between the roads.

To better understand the advantages and disadvantages of various models, we plot the prediction curves of some models

TABLE IV: Performance of MetaSTC and Baselines.

	Model	MAE ↓		MSE ↓		RMSE ↓	
		$L = 12$	$L = 24$	$L = 12$	$L = 24$	$L = 12$	$L = 24$
Beijing	ARIMA	10.607	10.757	177.101	185.531	13.308	13.633
	T-GCN	9.299	10.925	182.050	198.364	13.493	14.084
	TimesNet	3.903	4.253	36.351	40.874	6.029	6.393
	DLinear	4.294	4.180	38.846	38.287	6.233	6.188
	LSTM	4.837	4.844	46.483	46.472	6.818	6.817
	FiLM	3.391	3.547	28.226	29.663	5.313	5.446
	MetaSTC+LSTM	3.534	3.710	27.433	29.040	5.238	5.343
	MetaSTC+FiLM	3.367	3.476	26.893	27.527	5.19	5.247
Shanghai	ARIMA	9.026	9.303	134.129	142.827	11.581	11.951
	T-GCN	4.340	4.365	40.692	41.525	6.380	6.444
	TimesNet	4.338	4.787	44.385	51.402	6.662	7.170
	DLinear	4.560	4.364	46.176	43.031	6.795	6.560
	LSTM	5.816	6.167	67.160	80.620	8.195	8.980
	FiLM	4.389	4.215	45.241	39.901	6.726	6.317
	MetaSTC+LSTM	4.524	4.429	42.380	40.276	6.510	6.346
	MetaSTC+FiLM	4.018	4.173	37.076	37.992	6.089	6.164
PeMS04	ARIMA	10.644	10.661	160.730	163.333	12.678	12.780
	T-GCN	1.973	1.384	4.309	9.840	1.807	2.145
	Timesnet	1.114	1.574	4.667	9.354	2.160	1.789
	DLinear	1.197	1.388	4.573	4.998	2.138	2.236
	FiLM	0.947	1.140	3.199	4.573	1.789	2.148
	MetaSTC+LSTM	1.011	1.103	3.888	4.759	1.972	2.181
	MetaSTC+FiLM	0.870	0.859	2.589	2.360	1.609	1.536
LargeST	ARIMA	10.644	10.661	160.730	163.333	12.678	12.780
	T-GCN	6.159	5.294	64.770	55.042	8.048	7.419
	TimesNet	4.729	5.089	49.054	53.283	7.004	7.299
	DLinear	4.975	4.798	50.074	47.204	7.076	6.871
	LSTM	5.934	6.142	70.606	75.824	8.403	8.708
	FiLM	4.395	4.788	43.472	48.743	6.593	6.982
	MetaSTC+LSTM	4.644	5.032	45.520	49.102	6.747	7.007
	MetaSTC+FiLM	4.369	4.491	43.333	44.398	6.583	6.663

as shown in Figure 6. Among them, Meta-STC+FiLM achieves the best prediction performance. The ARIMA model has a relatively large prediction error, but it captures the long-term trend. The T-GCN model has the largest prediction error, this suggests that sparse data has the most severe effect on graph neural networks.

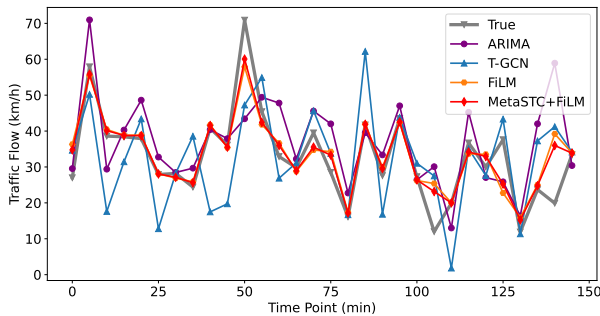


Fig. 6: Prediction Curves

C. Ablation Study: RQ2

In this section, we perform ablation studies to analyze the influence of key components within MetaSTC. The LSTM model is used to make the model lightweight and fast trainable. The experimental results are recorded in Figure 7. We

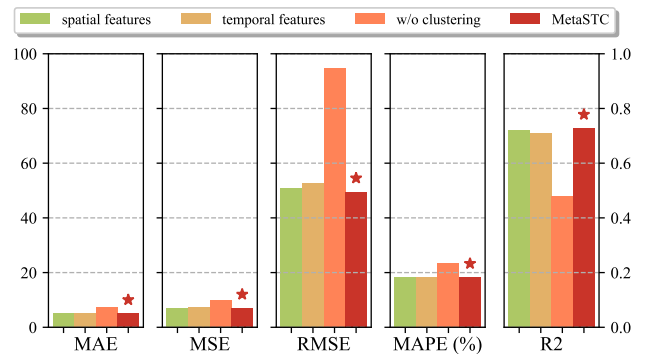


Fig. 7: Prediction Accuracy of MetaSTC and Ablation Versions of Major Parts in the Model

conducted a comparison between our complete MetaSTC model and two ablated versions to assess their respective performances. The first ablated version lacked the meta-learning component (in yellow), while the second one did not incorporate the clustering module (in green).

- **Effect of Clustering Module:** We observed that the clustering-based model outperformed the direct prediction model, with a 48.2% reduction in MAE error. By partitioning the data into different clusters, the model

can filter out outliers that do not belong to a particular category and focus on the features of a specific category, thus more accurately capturing the intrinsic patterns of the data in that category. In part IV-E we further analyzed and interpreted the results of the clustering.

- **Effect of Meta-Learning Module:** The experiment results indicated that the inclusion of meta-learning had a significant positive impact on the model’s performance, resulting in a 32.9% reduction in MAE error compared to the ablated version. By accumulating experience on multiple tasks, meta-learner allows our model to perform better on unseen data. We also show in part IV-E that the models that incorporate our framework converge faster (training time).

D. Interpretability Analysis: RQ2

To visualize the effectiveness of our clustering module, we randomly selected two clusters and then sampled 30 roads in each of them. As is depicted in the figure 8, the Pearson coefficients within clusters are neatly organized than between clusters (overall high or low). The Pearson coefficients of the road sequences within the first clusters are higher, which proves that they have more temporal similarity. The pearson coefficients within the second clusters are lower, which means that their similarity is mainly in spatial characteristics.

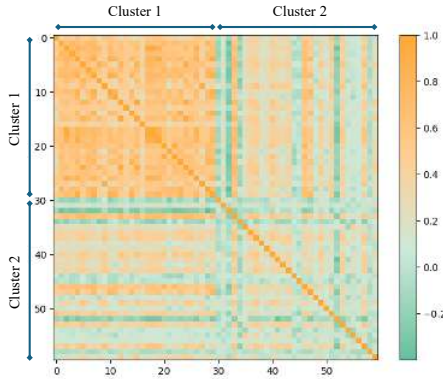


Fig. 8: Pearson Coefficients of Traffic Flow Sequences

In order to further analyze the results of clustering, we do the statistics of various spatial and temporal features of different clusters. The figure 9 shows the distribution of temporal features (e.g. number of lanes shown in Figure 9a) and the spatial features (e.g. average traffic flow shown in Figure 9b) for our five obtained clusters. First, the intra-cluster features have similarities and the inter-cluster features have significant differences in distribution. This indicates that the clustering algorithm may be effective in grouping data points with similar attributes together. These clustering results have some degree of internal consistency. Models tend to cluster areas with similar roadway structures or designs. The similarity in average flows reveals that the model also does a good job of aggregating different traffic patterns.

At the same time, we note the tendency to aggregate lower lane count lane counts, such as the third cluster in the Figure 9, which also aggregates to lane couples with lower average

flows. This may indicate that there is some match between roadway design (fewer lanes) and traffic demand (lower flows) in certain areas. Fewer lanes may be sufficient to meet traffic flow demands in these areas. This is in line with our usual perceptions and also shows that there will be some correlation between spatial and temporal features.

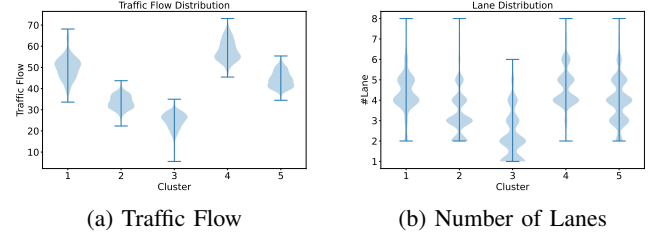


Fig. 9: Clustering Results

E. Comparative Results: RQ3

To show the effectiveness of our models in lightweight time series prediction models, we use the monetary cost of renting training GPUs and the time cost consumption as a measure. We use the GPU rental cost of 0.4 dollars per hour as a standard to estimate the training budget overhead of 500,000 roads per city. The cost consumption comparison of different models is shown in Table V.

TABLE V: Hourly Cost Estimation With and Without MetaSTC per City.

Model	Beijing	Shanghai	California
LSTM	3.59	3.24	2.26
MetaSTC+LSTM	1.90 (47.0%↓)	0.25 (92.3%↓)	0.19 (91.7%↓)
FiLM	20.35	21.83	13.58
MetaSTC+FiLM	14.66 (30.0%↓)	2.24 (89.7%↓)	1.59 (88.2%↓)

Besides, in scenarios where efficiency is more emphasized, we can combine the MetaSTC model with a more concise deep learning model such as LSTM to achieve a shorter training time. This demonstrates the advantages of MetaSTC in real-world applications. We can consider which prediction model to combine the MetaSTC model with in conjunction with the scenario’s requirements on accuracy and cost.

F. Journal Extension Evaluation: RQ4–RQ5

Multi-seed performance closure. Table VI compares the published ICDM 2024 MetaSTC result, a stronger static reproduction, and V14. All V14 numbers are mean $\pm$ standard deviation over seeds 42–46. The five-seed mean improves both published MAE and MSE on all three datasets, and all 15 dataset-seed MAE results beat the corresponding published value. Relative to the stronger internal static reproduction, V14 further improves Shanghai and LargeST, while Beijing is 0.34% worse in MAE. We therefore use the published conference result as the journal-extension reference and report the stronger reproduction transparently rather than claiming universal improvement over every internal static baseline.

TABLE VI: Five-seed journal evaluation under the official LSTM $L = 12$ protocol. Lower is better. “Wins” counts seeds that improve the published MetaSTC target.

Dataset	Published MetaSTC (MAE/MSE)	Strong Static (MAE/MSE)	V14 MAE	V14 MSE	Improvement (MAE/MSE)	Wins (MAE/MSE)
Beijing	3.534 / 27.433	3.4990 / 27.1918	3.5109 $\pm$ 0.0191	27.2152 $\pm$ 0.2567	0.654% / 0.794%	5/5 / 4/5
Shanghai	4.524 / 42.380	4.3744 / 40.7825	4.2654 $\pm$ 0.0177	39.3072 $\pm$ 0.2180	5.717% / 7.251%	5/5 / 5/5
LargeST	4.644 / 45.520	4.6162 / 45.9753	4.4582 $\pm$ 0.0328	44.6292 $\pm$ 0.4482	4.001% / 1.957%	5/5 / 5/5

TABLE VII: Matched-control analysis (five seeds). Each entry reports MAE / MSE; lower is better.

Variant	Beijing	Shanghai	LargeST
V14 context + task + gate	3.5109 $\pm$ 0.0191 / 27.2152 $\pm$ 0.2567	4.2654 $\pm$ 0.0177 / 39.3072 $\pm$ 0.2180	4.4582 $\pm$ 0.0328 / 44.6292 $\pm$ 0.4482
Constant gate	3.5023 $\pm$ 0.0174 / 27.1688 $\pm$ 0.1900	4.3130 $\pm$ 0.0065 / 39.9175 $\pm$ 0.0815	4.4746 $\pm$ 0.0114 / 44.5837 $\pm$ 0.1383
Shared, no task identity	3.5644 $\pm$ 0.0071 / 27.7553 $\pm$ 0.0429	4.2748 $\pm$ 0.0175 / 39.4279 $\pm$ 0.2182	4.3988 $\pm$ 0.0125 / 43.8861 $\pm$ 0.2401
Cluster-wise linear residual	3.4693 $\pm$ 0.0005 / 26.7278 $\pm$ 0.0083	4.3403 $\pm$ 0.0000 / 40.2853 $\pm$ 0.0000	4.5810 $\pm$ 0.0011 / 45.6352 $\pm$ 0.0035

TABLE VIII: Additional parameter footprint and batched inference cost of V14.

Dataset	Static Params	Added Params	Static Time (ms)	Full V14 Time (ms)
Beijing	16,430	10,871	2614.3 $\pm$ 150.6	2587.6 $\pm$ 133.8
Shanghai	9,858	10,855	42.4 $\pm$ 37.5	44.8 $\pm$ 37.7
LargeST	9,858	10,855	1212.1 $\pm$ 37.6	1207.4 $\pm$ 39.9

Matched residual controls. We compare V14 with a constant-gate adapter, a shared adapter without task identity, and a cluster-wise linear residual calibrator under the same protocol. Table VII shows that the richer mechanism is not uniformly necessary: V14 is best on Shanghai, the linear control is strongest on Beijing, and the shared-no-task control is strongest on LargeST. Residual adaptation reveals correctable dynamic bias in the static formulation, but the best adaptation form is dataset-dependent.

Efficiency. V14 contains about 10.9K trainable parameters. Since the LSTM anchors are themselves very small, this is not negligible as a percentage of the expert ensemble; we report the absolute footprint. In batched RTX 4090 evaluation, V14 is statistically indistinguishable from the static path on Beijing and LargeST and differs by only 2.4 ms over the full Shanghai test pass, with variance larger than that difference. Thus V14 adds no measurable material inference-latency penalty under this setting.

When does richer adaptation help? Shanghai is the clearest segment-level case. Across all 40 test windows, V14 improves static MAE by 0.0997, compared with 0.0341 for the linear residual control, and V14 wins every window. In the highest static-error tercile (13 windows), the V14 gain rises to 0.1349 and its additional gain over linear increases from 0.0657 to 0.0928. Static-anchor error strongly correlates with V14 gain (Pearson $r \approx 0.817$), and the gate positively correlates with realized gain ($r \approx 0.674$). Figure 10 summarizes this contrast.

This pattern is not universal. On Beijing the linear residual control is stronger overall; on LargeST removing explicit task identity gives the best average result. These counterexamples show that richer context conditioning is most valuable in particular dynamic regimes, while simpler calibration can be sufficient elsewhere.

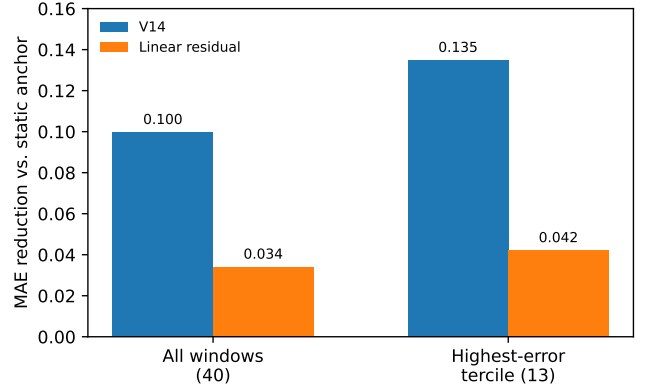


Fig. 10: Shanghai segment case study. Values are MAE reductions relative to the frozen static anchor; larger is better.

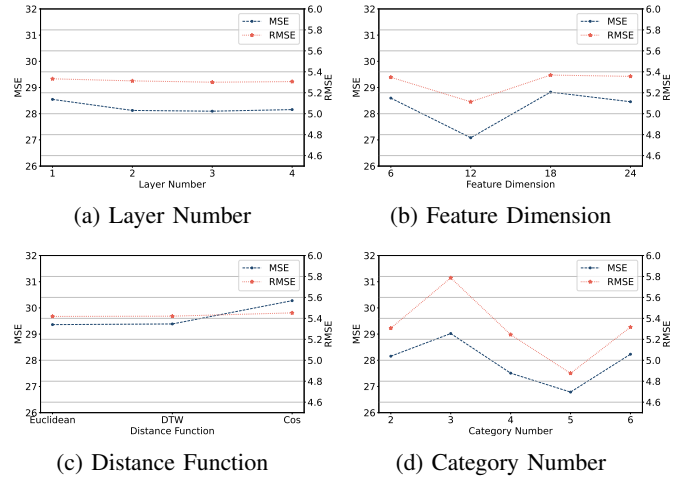


Fig. 11: Parameter Study.

G. Parameter Study: RQ3

We test the effect of the dimensionality reduction of temporal features, clustering distance functions, clustering categories, and the number of layers in the prediction meta-learner. We tried different parameter settings as is shown in Figure 11. The choice of the number of layers and the choice of the distance equation have a marginal effect on the result, however, the choice of the dimensionality of the features and the number

of clusters have a significant effect on the accuracy.

V. DISCUSSION

A. Application Potential and Practical Implications

Beyond addressing technical challenges, MetaSTC offers substantial potential for real-world traffic management. One key application lies in congestion prediction and mitigation. For example, our clustering module (Fig. 9) identifies distinct traffic patterns, such as Cluster 1 with high average flows during peak hours, indicative of urban bottlenecks (e.g., business districts in Beijing). By integrating MetaSTC's predictions, traffic authorities could dynamically adjust signal timings—extending green phases for high-flow clusters during rush hours—reducing delays based on baseline flow reductions observed in our LargeST dataset (Table IV). This targeted optimization leverages the model's ability to isolate critical zones, enhancing urban mobility without requiring extensive infrastructure upgrades.

Real-time applicability is equally critical, as traffic systems often demand instantaneous forecasting. MetaSTC's lightweight design, evidenced by reduced training times (Table V), supports online learning scenarios. Algorithm 1, which rapidly encodes new roads into clusters via spatio-temporal embeddings, enables incremental updates without retraining the entire model. For instance, adding a newly monitored road in Shanghai could be processed in seconds by assigning it to an existing cluster and refining local parameters, maintaining prediction accuracy under dynamic conditions. This capability positions MetaSTC as a viable tool for adaptive traffic control systems, such as real-time rerouting during unexpected events like accidents or road closures.

B. Limitations and Future Directions

The journal evidence exposes clear limitations. First, full V14 conditioning is not the best matched residual form on every dataset, so we do not claim a universal dynamic-task-discovery mechanism. Second, the present closure deliberately preserves the official LSTM $L = 12$ protocol for direct comparability with the conference result; architecture-generalization experiments are left to future work. Third, recent-history statistics are used as diagnostic proxies rather than as a new distribution-shift benchmark. Future work should study explicit temporal distribution shift, cross-city generalization, and test-time adaptation under separately frozen protocols rather than tuning these extensions post hoc to the present table.

VI. RELATED WORKS

Traffic Flow Prediction In the early days of traffic flow prediction research, researchers usually used traditional statistical methods for time-series prediction, such as ARIMA [2]. After that deep learning models were adopted as the amount of data increased. Convolutional Neural Networks (CNN) have given rise to several excellent network models [22]. Besides, transformer [20] and its variants [12] [5] [10] are able to automatically learn and focus on the important parts of the

input sequence for better understanding and processing of the sequence data through the attention mechanism.

With the rise of Graph Neural Network (GNN), there are a lot of GNN-based methods. They generally combine GNNs with either Recurrent Neural Networks (RNNs) or Temporal Convolutional Networks (TCNs) to capture intricate spatial and temporal dependencies in traffic data. ST-MetaNet [17] and AGCRN [1] employ RNNs and their variants. ASTCN [27], STGCN [24] and GWN [23] use dilated causal convolution for temporal patterns modeling. These approaches have exhibited promising performance on small-scale datasets, which, however, have limited size in terms of the number of nodes, edges, and time frames. This raises valid concerns about their scalability when applied to large-scale traffic forecasting datasets, which strive to encompass more realistic real-world scenarios.

Clustering in Traffic Flow Analysis Clustering algorithms have been widely used in the field of traffic flow analysis to improve the efficiency of traffic prediction problems. Clustering methods were first employed by Kuo et al. to improve traffic prediction [11]. After that, Tsai et al. developed a dual-module architecture comprising a Temporal Neighborhood Clustering module and an Incremental Learning module [19]. Luo et al. demonstrated that clustering could be a promising solution to the smoothing problem of GCN networks [15].

However, these methods often rely on a single feature of traffic flow data while neglecting the intersectionality between different features. In contrast to these studies, our clustering method focuses on the interaction between spatial and temporal features and designs a gating mechanism for extracting the spatio-temporal features so that we can cluster together roads with similar patterns to avoid unevenness in prediction accuracy among roads.

Meta-Learning in Traffic Flow Prediction Meta-learning is a machine-learning technique that aims to enable models to initialize parameters from the retrained models and adapt quickly to new environments or tasks. In recent years, meta-learning has become increasingly prevalent in predicting traffic flows. One widely used method involves extracting spatial information to initialize the parameters of the meta-model. Zhang et al. proposed an efficient meta-learning way that tunes the global model with specific heterogeneous information from different regions [26]. Besides, other features have also been used for meta-learning. For instance, Pan et al. incorporate two meta-learners to assist the predictions of the main model [17]. Fang et al. fused two meta-learning modules that integrate temporal and spatial data dimensions [6]. In addition, Wang et al. conducted a study on graph structure in meta-learning [21]. Our model first combines clustering results and meta-learning techniques in traffic flow prediction. We also designed a meta-learning method to incorporate context information into the meta-learning model.

Dynamic Heterogeneity and Residual Adaptation Existing traffic meta-learning work primarily models heterogeneity through spatial structures, task-specific parameters, or meta-learned initialization. Our journal extension addresses a complementary issue: after a road has a useful static task prior, its recent state may still differ from the regime represented by

the static expert. We therefore preserve the static expert and represent current state continuously from recent observations and the expert forecast. The residual form separates whether the conference predictor should remain trusted as an anchor from whether the current context justifies a bounded correction. The matched controls in Sec. IV-F show that this richer conditioning adds value in some settings without implying that every traffic regime requires the most expressive adapter.

VII. CONCLUSION

We presented a journal extension of MetaSTC for large-scale traffic forecasting under both persistent and time-varying spatio-temporal heterogeneity. The original clustering and meta-learning framework is retained as a strong static anchor. A context-conditioned gated residual meta-adapter summarizes recent history and the static forecast into a continuous state, combines it with the static task prior, and learns only a bounded correction. Zero initialization and a safety-aware objective allow conservative inheritance of the conference solution. Under the official LSTM $L = 12$ protocol, the five-seed V14 mean improves the published MetaSTC MAE and MSE on Beijing, Shanghai, and LargeST, with all 15 dataset-seed MAE results beating the published targets. The adapter has a small absolute footprint and no measurable material inference-latency penalty under batched GPU evaluation. Matched controls also show that the richest context/task-conditioned mechanism is not universally optimal. We therefore interpret the result as setting-dependent adaptive correction on top of a reliable static model, and report both the positive Shanghai hard-window case and contrasting datasets to define the boundary of the claim.

VIII. ACKNOWLEDGEMENTS

We are deeply grateful to all who have contributed to this paper. Special thanks go to Kexin Xu and Zhemeng Yu. Kexin Xu provided research thinking with profound expertise. Zhemeng Yu dedicated great efforts to data collection and analysis, laying a solid foundation for the empirical part of the paper.

This work was supported by the National Key R&D Program of China [2024YFF0617700], the National Natural Science Foundation of China [U23A20309, 62272302, 62372296], and the CCF-DiDi GAIA Collaborative Research Funds for Young Scholars [202404].

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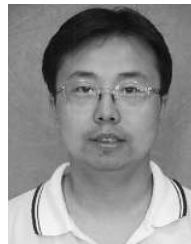


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